From Defect Detection to Flight Clearance: a Bayesian SHM Stack for Reusable Launch Vehicles

Stage 0–3 implemented and verified on CFRP thermoelastic fields

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June 2026

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Motivation: reusable rockets change the question

Expendable: "will it survive *one* flight?" — a single pass/fail margin check.

Reusable: "is this vehicle OK to fly *again*, and how many flights remain?" — a recurring, history-dependent decision.

- Same vehicle re-inspected after every flight ⇒ *fixed geometry, fixed individual* is the norm, and its own past-flight scans form a growing healthy reference
- SpaceX (empirically): preventive replacement → *condition-based monitoring*; staged certification 20 → 40 flights; fleet-leader B1067 at 35 flights (06/2026)
- Cost asymmetry: vehicle + payload loss ≫

This work

Formalise that practice as a *measurable, Bayesian, physics-based* pipeline — detect → characterise (with uncertainty) → predict crack growth → clear — applicable from the *first* article, not after a decade of flights.

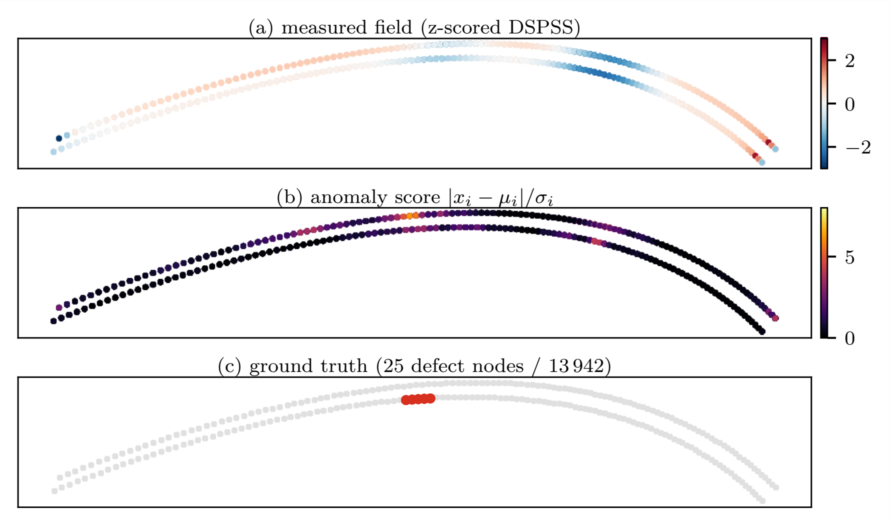
Data: CFRP interstage thermoelastic full-field (DSPSS, 13 942 nodes) + H3 fairing guided waves.

The Stage 0–5 stack

Stage 0 Screening Mahalanobis → **Stage 1** Classification GNN → **Stage 2** Characterisation FMPE posterior → **Stage 3** Prognosis phase-field + fatigue → **Stage 4–5** Fleet learning design feedback

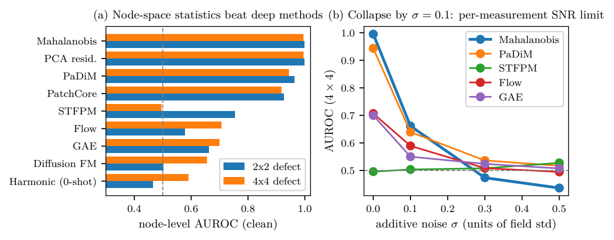
- **Green = implemented and tested** (2026-06; two git repos, all committed; 19/19 tests green)
- Each stage answers one question: **where?** (Stage 0) → **what?** (Stage 1) → **how bad, with what uncertainty?** (Stage 2) → **fly again?** (Stage 3)
- One data contract end-to-end: posterior $\theta = (c_x, c_y, \text{layer}, \log_2 \text{size})$ flows from Stage 2 *directly* into Stage 3's defect seeding
- Stages 0–2 are model-light (statistics / amortised inference); Stage 3 carries the mechanics (phase-field fracture)

What the system sees: detection at a glance



The measured field (a) looks unremarkable; the per-node score (b) lights up exactly where the 25 defect nodes sit

Stage 0 — a 30-line statistic beats eight learned methods



Score $s_i = |x_i - \mu_i|/\sigma_i$, with (μ_i, σ_i) per node from N=2000 healthy fields. **Node AUROC (clean):**

- Mahalanobis / PCA: **0.999 / 0.995** (2x2 / 4x4)
- PaDiM 0.96, PatchCore 0.93, STFPM 0.75, GAE 0.66, diffusion 0.50–0.66

SNR limit — measured, not folklore

On a labelled coupon, detection AUROC is still $\sim$0.84 at σ=0.1 of the field std and only reaches chance near $\sigma \approx$0.7–1.0 — *refuting* the "collapse at 0.1" folklore. Lock-in IR ($\sigma \propto 1/\sqrt{K}$) extends it further: an SNR *requirement*, not a blocker.

Why the simple method wins here — and why GNN still matters

Fixed geometry = template statistics

Re-inspecting the *same* vehicle: healthy manifold is a point cloud around one template. Per-node (μ_i, σ_i) is the correct inductive bias.

Complementary weaknesses

	Mahalanobis	GNN
unknown defect size	open	collapses
noise (12%)	dies	robust
semantic class	—	only option
new airframe	needs refs	transferable

$\Rightarrow$ statistics dominate as a vehicle accumulates flight history; learned models bootstrap *new* airframes and repairs.

Same geometry-only Tier-B descriptor, **leave-one-structure-out** over 7 real structures (5 modalities: DSPSS / DIC / guided-wave / DFOS / vibration).

held-out structure	single cue	learned
DIC panel (crack)	$+0.71$	$+0.66$
omega-stringer (debond)	$+0.50$	$+0.50$
COPV membrane (burst)	-0.83	$+0.85$
interstage (1-node defect)	$+0.00$	-0.02
stiffened panel (load-shed)	-0.02	-0.15
Hawk *vibration* (mass)	-0.25	-0.75
dense guided-wave (hole)	$+0.38$	-0.04

Learning crosses *topology*

The COPV membrane inversion ($-0.83 \rightarrow +0.85$, perm. $p<0.001$) was an artefact of the naive cue — a richer descriptor reads the higher-order distribution shape.

Three boundaries survive learning

scale — a 1-node defect is invisible to any global descriptor; only the reference-based **Mahalanobis** node residual recovers it (**node-AUROC 0.9999**).

modality sign — vibration mass-loading *smooths* the field $\Rightarrow$ strongly informative but sign-inverted (-0.75).

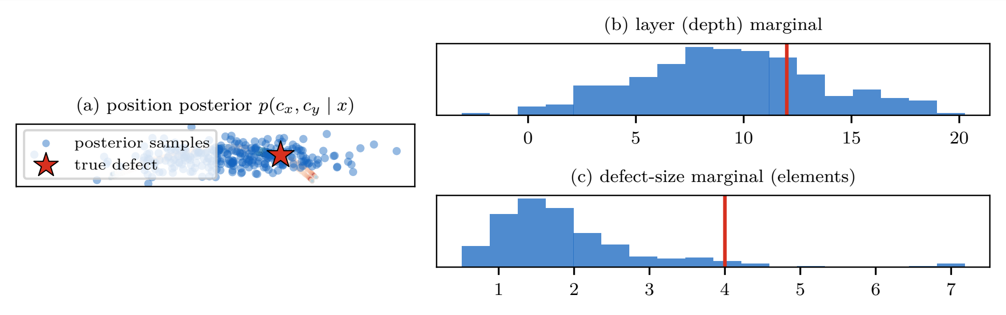
weak scatterer — a small guided-wave hole is too diffuse to transfer ($\rho \approx 0$).

Stage 2 — calibrated Bayesian defect characterisation (FMPE)

Flow-matching posterior estimation, trained on the *existing* FEM set (20k pairs, zero new simulations):

parameter	RMSE	R^2	90%-coverage
position c_x	0.055	0.78	**0.944**
position c_y	0.021	0.83	**0.940**
size (3-class)	—	0.73 (acc. 0.82)	0.894
layer (depth)	2.98	0.63	0.870

- Coverage $\approx$ target 0.90 $\Rightarrow$ **calibrated** posteriors, safe to propagate downstream
- Depth weakly identifiable — a physics limit of surface thermoelastic measurement, reported honestly



One measured field → 256 posterior draws. Position (a) concentrates at the true defect (star); depth (b) is honestly broad — the physics limit of a surface measurement; size (c) shows why we propagate the *whole* posterior into Stage 3 instead of trusting one number.

Stage 3 — phase-field prognosis with true fatigue

Energy $\Psi = \int \left[g(d)\,\psi_e^+ + \frac{G_c}{c_w}\left(\frac{d^2}{\ell} + \ell\,\nabla d\cdot A\,\nabla d\right)\right] dV$, with structural tensor $A = I + \beta\, a\otimes a$ (fiber dir. a).

Model (12/12 tests green):

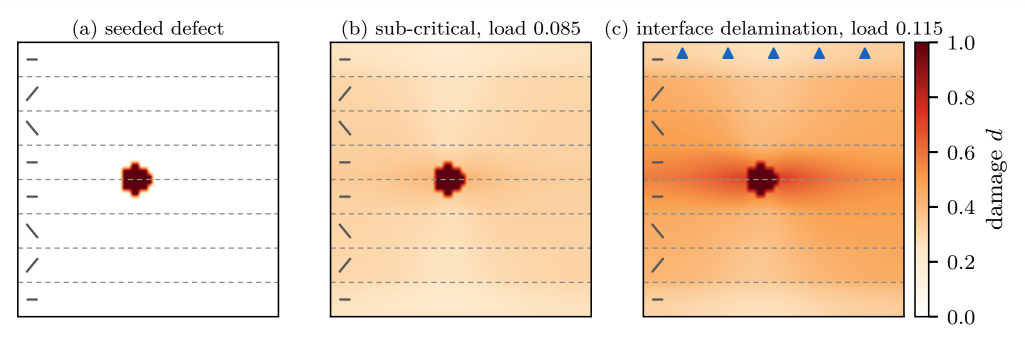
- multi-ply: per-ply $A(\theta_{\mathrm{ply}})$, horizontal bands
- interface bands G_c^{int}/G_c^{ply}=0.4 (DCB/ENF-anchored) — pure-PF surrogate of Paggi–Reinoso CZM
- Carrara (2020) fatigue: history $\bar{\alpha}$ accumulates ψ_e^+ over cycles, $f(\bar{\alpha})$ degrades $G_c \Rightarrow$ Wöhler-like growth under *repeated sub-critical* loads

Verified physics:

- fiber at 30° $\rightarrow$ crack deflects to 23.3°
- low-G_c interface captures crack (73 vs 0 new cells)
- P(growth) monotone in load: $0 \rightarrow \frac{2}{3} \rightarrow 1$
- FD, plane-strain, 60×48 grid, ~15 s/run

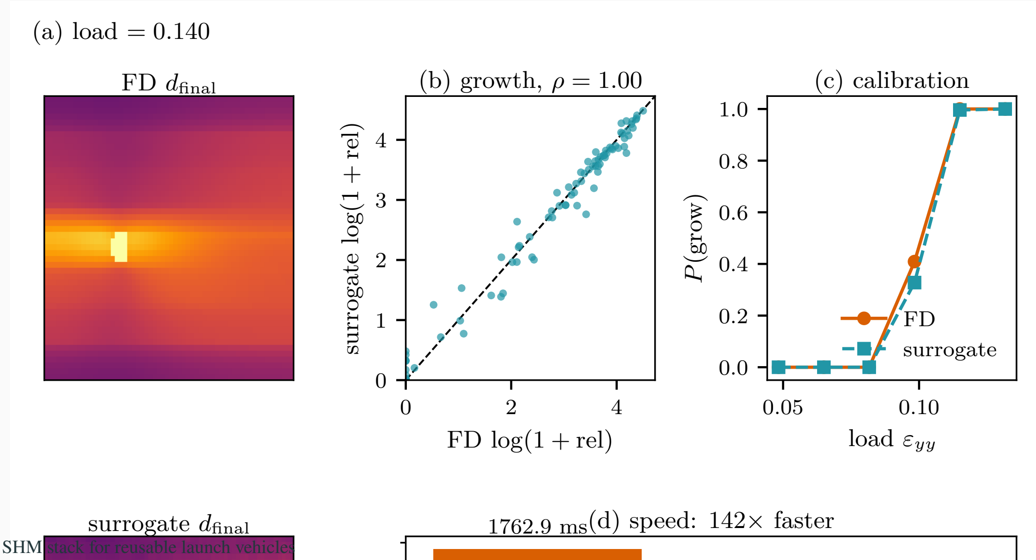
Caveat: load is a normalised Mode-I peel proxy; $\bar{\alpha}_T$ not yet S–N-anchored.

Stage 3 in pictures: the posterior defect grows — or doesn't

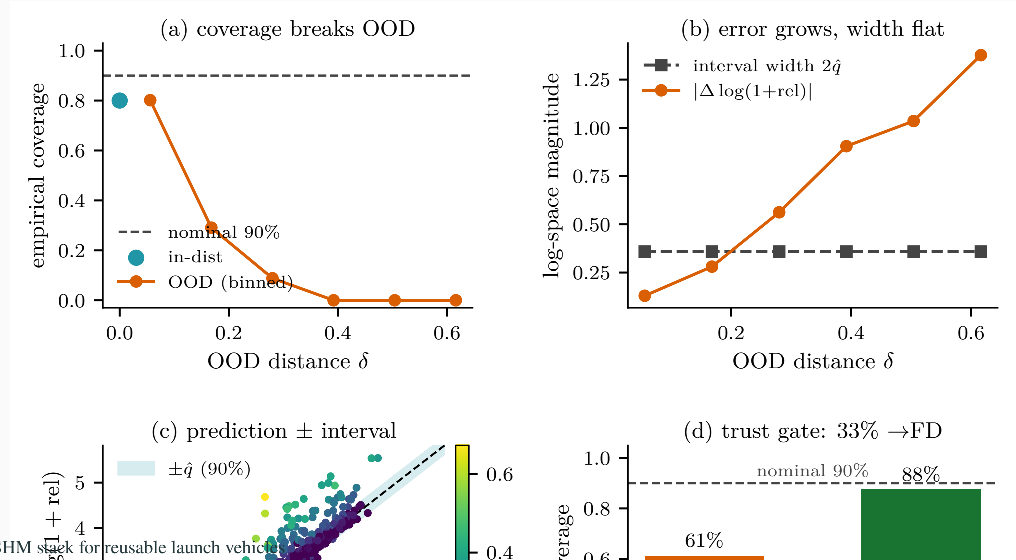


- (a) initial damage seeded at the FMPE posterior mean (c_x, c_y, layer, size); dashed lines = ply interfaces, glyphs = fiber angles $(0/\pm 30)_s$
- (b) below critical load: blunted, no propagation
- (c) above: the crack selects the weak interface path (delamination), exactly the failure mode DCB/ENF data anchor (G_c^{int}/G_c^{ply}=0.4); arrows = transverse tension

Stage 3, made fast: an FNO surrogate for fleet-scale use



Knowing when to trust it: conformal OOD trust-gate



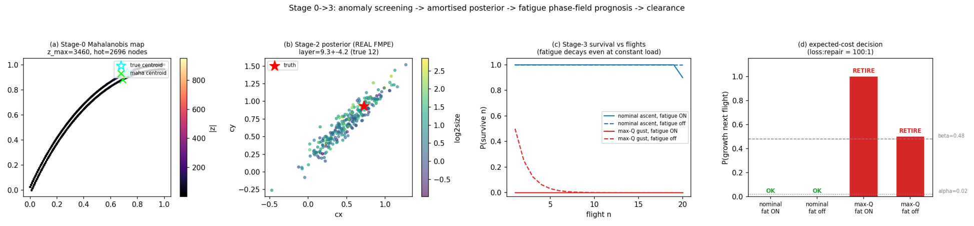
Flight clearance — end-to-end on a real test sample

Stage 0→3 run: held-out 4x4 field → Mahalanobis map → *trained* FMPE posterior → fatigue-aware clearance (20-flight horizon).

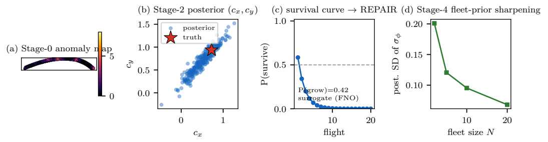
load case	load	fatigue	P(grow)	$\mathbb{E}$[flights]	decision
nominal ascent	0.06	**ON**	0.00	**19.9**	**OK**
nominal ascent	0.06	off	0.00	11.0	OK
max-Q gust	0.10	**ON**	1.00	0.0	**RETIRE**
max-Q gust	0.10	off	0.50	1.0	RETIRE

- fatigue *accumulates across flights* — P(survive 20) now decays even at constant sub-critical load (no i.i.d. shortcut)
- thresholds from expected cost (vehicle:repair = 100:1 ⇒ α=0.02, β=0.48)

End-to-end, one figure: detect → characterise → clear



Stage 0 anomaly map · Stage 2 posterior · $P(\text{survive } n)$ with fatigue · decision — single held-out CFRP field.

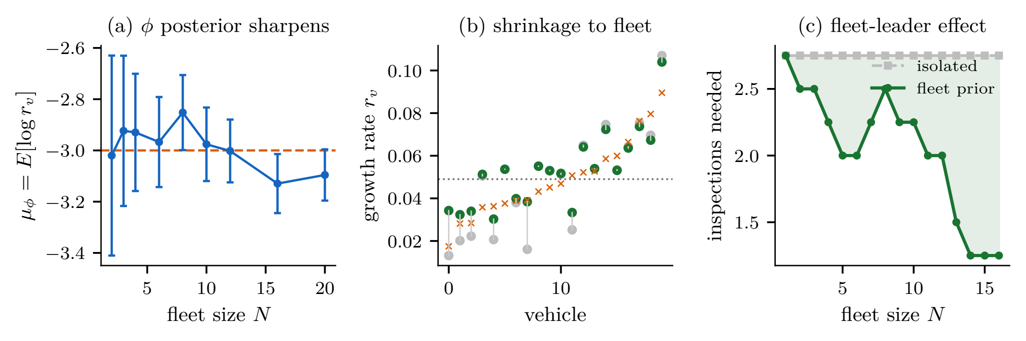



```
$ python run_pipeline.py -sample 3

[0 DETECT]   AUROC 0.998   → DEFECT
[2 CHARAC]   FMPE  cx 0.50±0.27 ...
[3 PROGNO]   P(grow)=0.42
             E[remain]=1.4   (FNO 37ms/draw)
[DECISION]   → REPAIR
[4 FLEET ]   prior SD 0.20→0.07
```

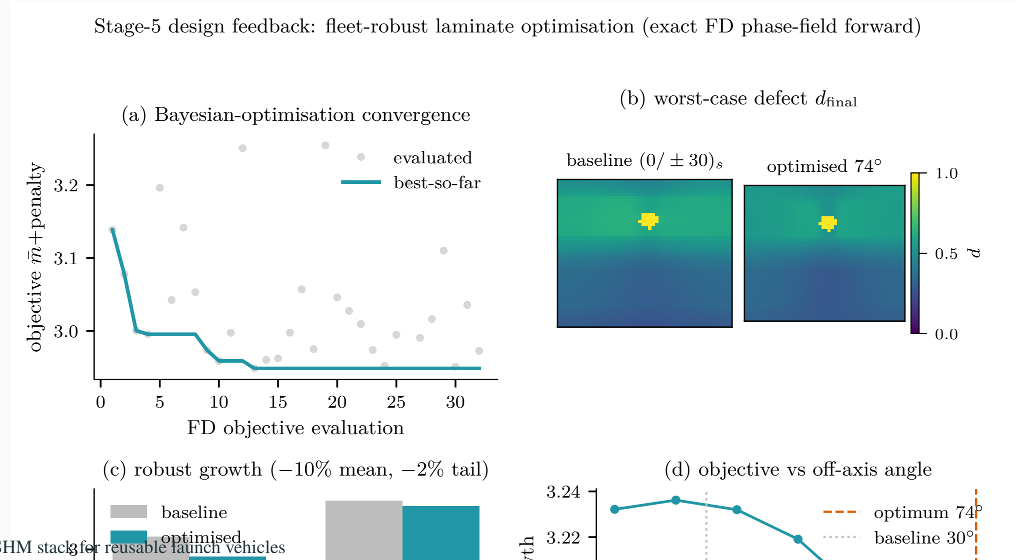
- every stage *imported unchanged*; surrogate engine ⟹ whole run <4 s (FD engine: Stage 3 alone ∼10 s)
- one reproducible artifact = paper + live demo + Keio proposal

Stage 4 — fleet learning: the SpaceX loop, formalised



- hierarchical Bayes: fleet prior $\varphi \rightarrow$ per-vehicle growth rate $s_v \rightarrow$ inspection posteriors; conjugate Gibbs
- (a) φ sharpens with flights (SD 0.39 → 0.10); (b) low-data vehicles shrink toward the fleet mean (error ratio 0.84)
- (c) a new vehicle clears in 1.25 inspections with the fleet prior vs 2.75 in isolation — the fleet-leader effect, quantified

Stage 5 — design feedback: the fleet reshapes the next vehicle



Is the final call right? End-to-end decision UQ

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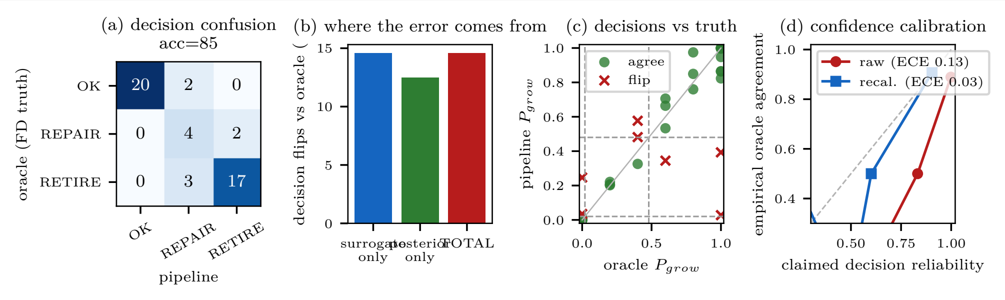
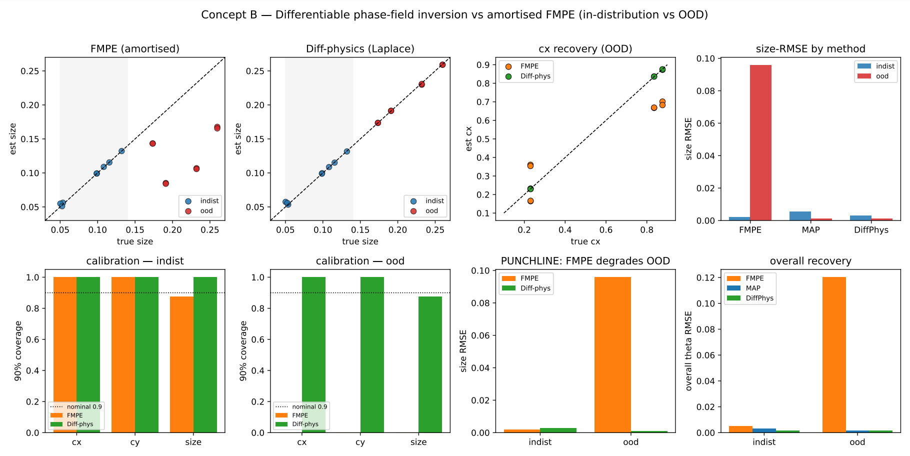


- every stage is calibrated alone — but the operator only sees the final OK/REPAIR/RETIRE. Score it against an *oracle*: exact FD over operational load scatter ⇒ true P_{grow} ⇒ oracle decision (no new data)
- 40 scenarios, real FMPE posteriors: 85% end-to-end accuracy; dangerous-miss $P(\text{OK} \mid \text{oracle=RETIRE}) = 0\%$ — all errors stay in the low-consequence OK↔REPAIR band
- decomposition: posterior 25% > surrogate 17.5%, partially cancelling to 15% total; per-decision confidence overconfident (ECE 0.18) → Platt recalibration (LOO) ECE 0.18 → 0.06, claimed reliability 96.5% → 84.5% vs empirical 85%

Closing the loop: differentiable physics vs amortised inference



- JAX-differentiable AT2 forward $F(\theta)$; gradient verified to rel. $3{\times}10^{-7}$; Laplace posterior from $J = \partial F/\partial\theta$, *no labelled pairs*
- OOD: FMPE degrades 25× coverage 0.96 → 0.00; the physics-grounded posterior is invariant (c RMSE

Outlook: Stage 4–5 and deployment

- **Stage 4 — fleet learning:** hierarchical Bayes over tail numbers; formalises the SpaceX fleet-leader practice (lead vehicle = likelihood, fleet = prior update)
- **Stage 5 — design feedback:** fleet damage statistics → layup/reinforcement optimisation (BO)
- **Sim-to-real:** flow-matching bridge ready for first measured full-field data (per-sample normalisation needs no healthy reference)
- **Targets:** JAXA reusable-vehicle programmes; journal paper (anomaly benchmark + calibrated FMPE + fatigue prognosis)

One sentence

SpaceX learned this loop empirically over a decade — we formalise it so it works from the first inspection of the first article.